

SQL Murder Mystery — Case Report

SQL City Police Department · Investigator: Adeking · Case Date: January 15, 2018

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1. The Case

On January 15, 2018, a murder occurred in SQL City. The original crime scene report has been lost, so this investigation begins by re-querying the police department's SQLite database (sql-murder-mystery.db) to rebuild the report, identify the witnesses, and trace the evidence back to a suspect and, ultimately, to whoever hired them.

2. Step 1 — Retrieve the Crime Scene Report

Query the `crime_scene_report` table for a murder that took place in SQL City on 2018-01-15 (stored as 20180115):

```
SELECT * FROM crime_scene_report
WHERE date = 20180115
      AND type = 'murder'
      AND city = 'SQL City';
```

Result: Security footage shows two witnesses. Witness #1 lives at the last house on "Northwestern Dr". Witness #2, named Annabel, lives somewhere on "Franklin Ave".

3. Step 2 — Identify the Witnesses

3a. Witness #1: last house on Northwestern Dr

"Last house on the street" means the highest street (house) number on Northwestern Dr:

```
SELECT * FROM person
WHERE address_street_name = 'Northwestern Dr'
ORDER BY address_number DESC
LIMIT 5;
```

Result: The highest address number belongs to Morty Schapiro (person id 14887) at 4919 Northwestern Dr.

3b. Witness #2: Annabel on Franklin Ave

```
SELECT * FROM person
WHERE address_street_name = 'Franklin Ave'
      AND name LIKE 'Annabel%';
```

Result: Annabel Miller (person id 16371) at 103 Franklin Ave.

4. Step 3 — Read the Witness Interviews

```
SELECT * FROM interview
WHERE person_id IN (14887, 16371);
```

Morty Schapiro (14887): "I heard a gunshot and then saw a man run out. He had a 'Get Fit Now Gym' bag. The membership number on the bag started with '48Z'. Only gold members have those bags. The man got into a car with a plate that included 'H42W'."

Annabel Miller (16371): "I saw the murder happen, and I recognized the killer from my gym when I was working out last week on January the 9th."

This gives three independent clues to triangulate the killer: (1) gym membership id starting with 48Z, (2) gold membership status, (3) a check-in at the gym on 2018-01-09, and (4) a license plate containing H42W.

5. Step 4 — Narrow Down the Suspect

5a. Gold members with an id starting with 48Z

```
SELECT * FROM get_fit_now_member
WHERE id LIKE '48Z%'
      AND membership_status = 'gold';
```

id	person_id	name	membership_start	status
48Z7A	28819	Joe Germuska	20160305	gold
48Z55	67318	Jeremy Bowers	20160101	gold

Two candidates remain. Check who among them checked in to the gym on January 9, 2018:

```
SELECT * FROM get_fit_now_check_in
WHERE check_in_date = 20180109
      AND membership_id IN ('48Z7A', '48Z55');
```

Both Joe Germuska (48Z7A) and Jeremy Bowers (48Z55) checked in that day, so the tie-breaker is the license plate containing 'H42W':

```
SELECT p.id, p.name, d.plate_number, d.car_make, d.car_model
FROM person p
JOIN drivers_license d ON p.license_id = d.id
WHERE p.id IN (28819, 67318);
```

Result: Only Jeremy Bowers has a matching plate **0H42W2** (Chevrolet Spark LS). Joe Germuska's plate does not contain "**H42W**".

Suspect identified: Jeremy Bowers (person id 67318).

6. Step 5 — But Was Bowers Acting Alone?

A good detective checks the suspect's own interview before closing the case:

```
SELECT * FROM interview
WHERE person_id = 67318;
```

Jeremy Bowers' statement: "I was hired by a woman with a lot of money. I don't know her name but I know she's around 5'5" (65") or 5'7" (67"). She has red hair and she drives a Tesla Model S. I know that she attended the SQL Symphony Concert 3 times in December 2017."

Bowers was a hired gun. The real mastermind still needs to be identified using her description: female, 65–67 inches tall, red hair, drives a Tesla Model S, and attended the SQL Symphony Concert three times in December 2017.

7. Step 6 — Identify the Mastermind

```
SELECT p.id, p.name
FROM person p
JOIN drivers_license d ON p.license_id = d.id
WHERE d.gender = 'female'
      AND d.height BETWEEN 65 AND 67
      AND d.hair_color = 'red'
```

```
AND d.car_make = 'Tesla'
AND d.car_model = 'Model S';
```

person_id	name
78881	Red Korb
90700	Regina George
99716	Miranda Priestly

Three people match the physical description and car. The concert attendance clue breaks the tie:

```
SELECT person_id, COUNT(*) AS visits
FROM facebook_event_checkin
WHERE event_name = 'SQL Symphony Concert'
      AND date LIKE '201712%'
      AND person_id IN (78881, 90700, 99716)
GROUP BY person_id
HAVING visits = 3;
```

Result: Only Miranda Priestly (person id 99716) attended the SQL Symphony Concert exactly 3 times in December 2017 (on Dec 6, Dec 12, and Dec 29).

8. Conclusion

The trail runs from the crime scene, through two witnesses, into gym records, vehicle registration, and finally a matching physical description and social check-in history. Every clue converges on a single person.

The mastermind behind the murder is Miranda Priestly.

Jeremy Bowers pulled the trigger, but he confirmed under questioning that he was hired by Miranda Priestly matched conclusively by her height, red hair, Tesla Model S, and her three December 2017 visits to the SQL Symphony Concert, a detail only the true mastermind's Facebook check-in history could confirm.

9. Investigative Method Summary

The case was solved by chaining foreign keys across the schema in this order:

crime_scene_report → person (witnesses) → interview → get_fit_now_member / get_fit_now_check_in → drivers_license → person (suspect) → interview (suspect's statement) → drivers_license / facebook_event_checkin → person (mastermind).

Each step used a specific, falsifiable clue (an address, a partial membership ID, a partial plate, a height range, a hair color, a car model, a concert-attendance count) to eliminate all but one match the hallmark of the relational-database approach to this mystery.

10. Addendum — Corroborating the Motive

The income table was not needed to identify the mastermind, but it independently corroborates Jeremy Bowers' statement that he was "hired by a woman with a lot of money":

```
SELECT p.name, i.annual_income
FROM person p
JOIN income i ON p.ssn = i.ssn
WHERE p.id IN (67318, 99716, 90700, 78881);
```

name	role in case	annual_income
Jeremy Bowers	hitman	\$10,500
Miranda Priestly	mastermind	\$310,000
Regina George	ruled out	no record
Red Korb	ruled out	\$278,000

Two things stand out. First, Miranda's income is roughly 30x Bowers' consistent with his description of an employer with significant wealth. Second, income alone would not have solved the case: Red Korb was almost as wealthy as Miranda, and Regina George had no income record to compare at all. The concert-attendance count (Section 7) remains the only clue that uniquely isolates Miranda from the other two physical-description matches income is corroborating evidence, not the deciding one.

Finally, it's worth noting a gap in the dataset: there is no interview record for Miranda Priestly herself. Her role as mastermind is established entirely through Bowers' testimony plus independent matches on physical description, vehicle, social check-in history, and now income not through any direct statement or confession of her own.